



Classification I: Basic Classifiers

Lecture 5

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Classification Problems

- Input: $\mathbf{x} = (x_1, x_2, \dots, x_d)$
- Predict the output (class label) $y \in \mathcal{Y}$
 - Binary classification: $\mathcal{Y} \in \{0, 1\}$
 - Multi-class classification: $\mathcal{Y} = \{1, 2, \dots, C\}$
- Regression: $f(\mathbf{x}) : \mathbb{R}^d \mapsto \mathbb{R}$
- **Learn a classification function:** $f(\mathbf{x}) : \mathbb{R}^d \mapsto \mathcal{Y}$

Classification

categorical
categorical
continuous
class

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Training set



Learn classifier



Model



Predicted classes

Different Classifiers



Refund	Marital Status	Taxable Income	Cheat
No	Single	75K	?
Yes	Married	50K	?
No	Married	150K	?
Yes	Divorced	90K	?
No	Single	40K	?
No	Married	80K	?

Test set





Outline

- K Nearest Neighbor Classifier
- Naïve Bayes
- Logistic Regression
- Case Study

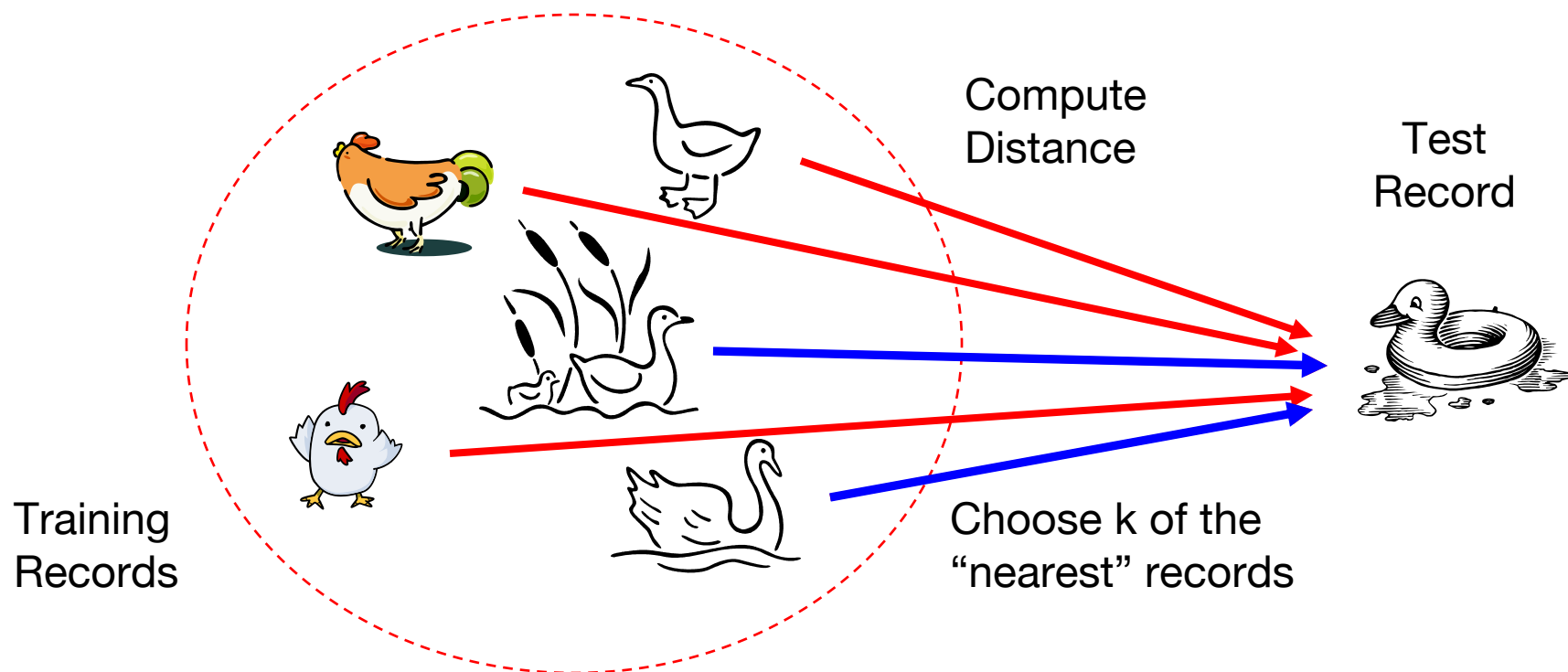


Outline

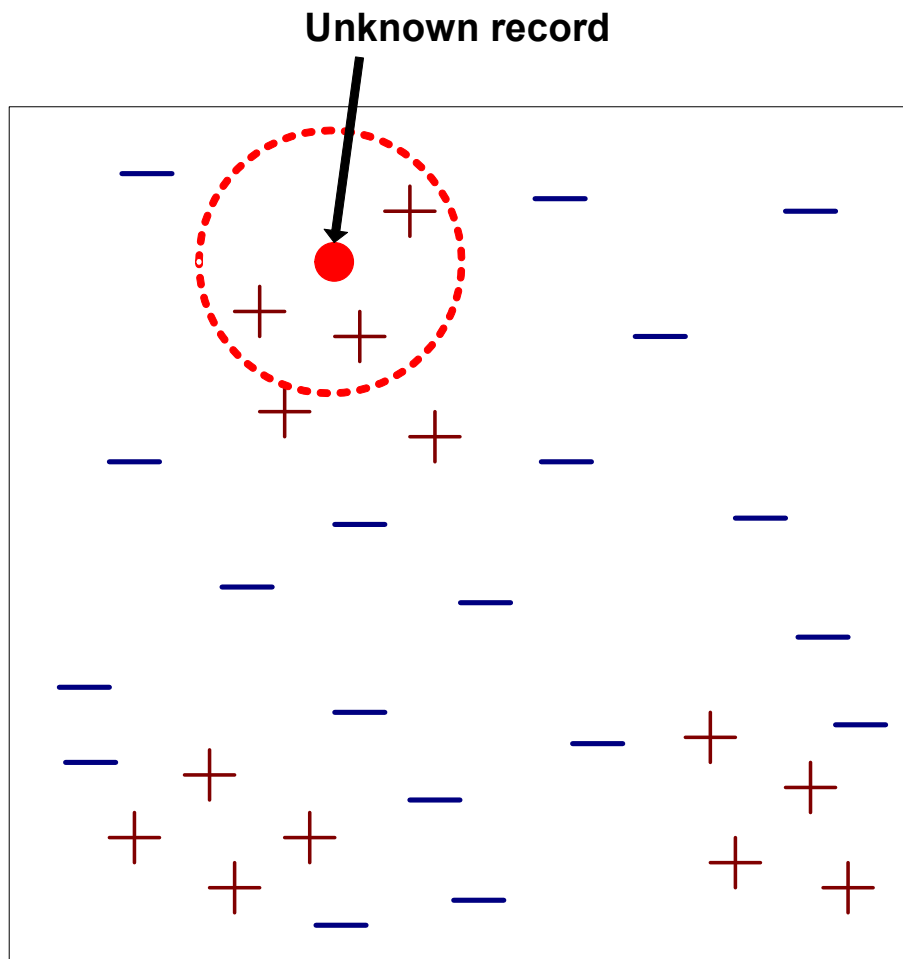
- **K Nearest Neighbor Classifier**
- Naïve Bayes
- Logistic Regression
- Case Study

K Nearest Neighbor Classifiers

- Basic idea:
 - If it walks like a duck, quacks like a duck, then it's probably a duck



K-Nearest-Neighbor Classifiers



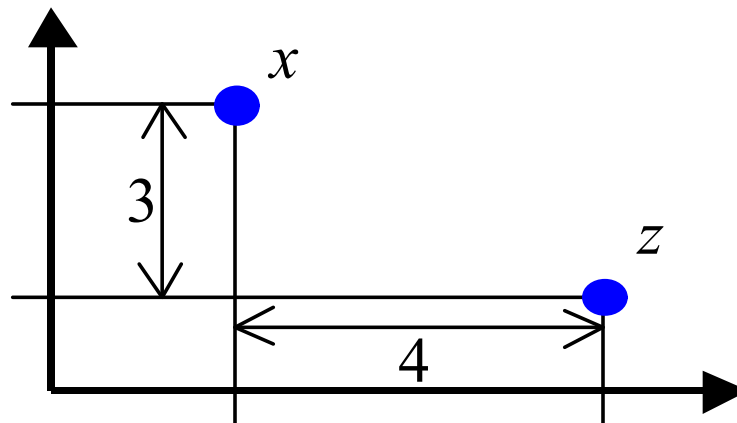
- | Requires three things
 - The set of stored records
 - Distance Metric to compute distance between records
 - The value of k , the number of nearest neighbors to retrieve
- | To classify an unknown record:
 - Compute distance to other training records
 - Identify k nearest neighbors
 - Use class labels of nearest neighbors to determine the class label of unknown record (e.g., by taking majority vote)

Distance metrics

- Compute distance between two points:
 - Euclidean distance
 - Manhattan distance
 - Sup-distance

$$x = (x_1, x_2, \dots, x_p)$$

$$z = (z_1, z_2, \dots, z_p)$$



Euclidean distance

$$d(x, z) = \sqrt{\sum_{i=1}^p |x_i - z_i|^2} \quad 5$$

Manhattan distance

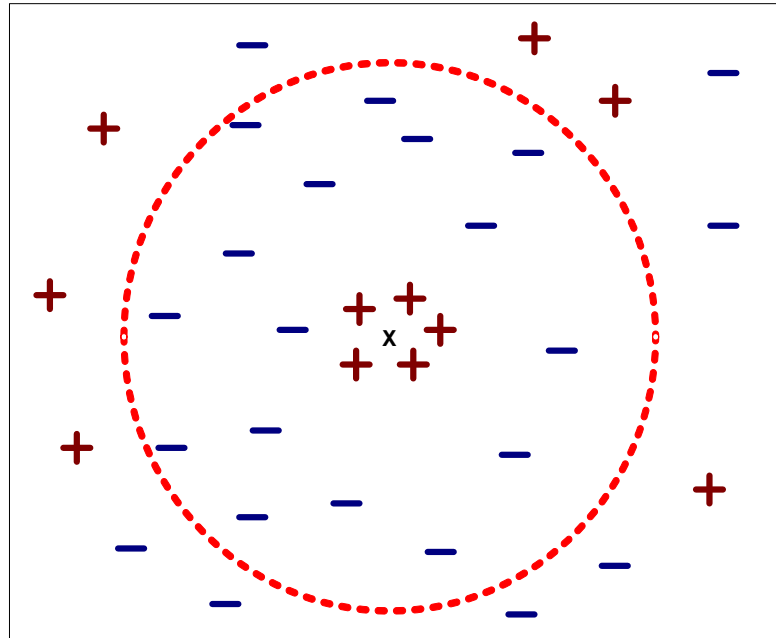
$$d(x, z) = \sum_{i=1}^p |x_i - z_i| \quad 7$$

Sup-distance

$$d(x, z) = \max_{1 \leq i \leq p} |x_i - z_i| \quad 4$$

K Nearest Neighbor

- Choosing the value of k :
 - If k is too small, sensitive to noise points
 - If k is too large, neighborhood may include points from other classes



Solution: K -cross
validation



Class Type Determination

- Determine the class from nearest neighbor list
 - Take the **majority vote** of class labels among the k-nearest neighbors
 - Weigh the vote according to distance
 - weight factor, $w = 1/d^2$
- Instance-based classifier



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Bayes' Rule

- Conditional probability: Event probability *given* that another event occurred

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- Bayes' rule (Relate $P(A|B)$ to $P(B|A)$)

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

Bayesian Learning

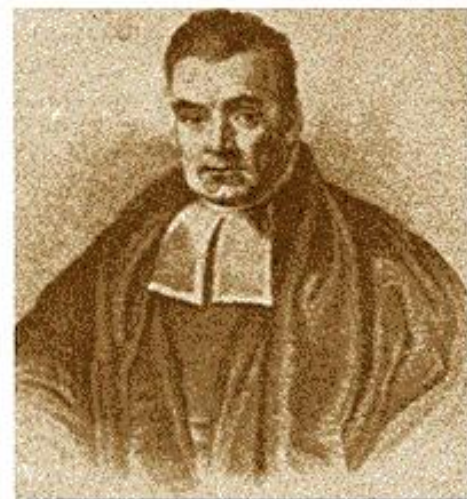
- Use Bayes rule:

$$P(\theta \mid \mathcal{D}) = \frac{P(\mathcal{D} \mid \theta)P(\theta)}{P(\mathcal{D})}$$

- Or equivalently:

$$P(\theta \mid \mathcal{D}) \propto P(\mathcal{D} \mid \theta)P(\theta)$$

posterior likelihood prior



Bayes, Thomas (1763) An essay towards solving a problem in the doctrine of chances. *Philosophical Transactions of the Royal Society of London*, 53:370-418

Bayesian Classifiers

- Consider each attribute and class label as random variables
- Given a record with attributes $(A_1, A_2, \dots, A_n)$
 - Goal is to predict class Y
 - Specifically, we want to find the value of Y that maximizes $P(Y | A_1, A_2, \dots, A_n)$
- Can we estimate $P(Y | A_1, A_2, \dots, A_n)$ directly from data?

Bayesian Classifiers

- Approach:
 - compute the posterior probability $P(Y \mid A_1, A_2, \dots, A_n)$ for all values of Y using the Bayes theorem

$$P(Y \mid A_1 A_2 \dots A_n) = \frac{P(A_1 A_2 \dots A_n \mid Y) P(Y)}{P(A_1 A_2 \dots A_n)}$$

- Choose value of Y that maximizes $P(Y \mid A_1, A_2, \dots, A_n)$
- Equivalent to choosing value of Y that maximizes $P(A_1, A_2, \dots, A_n \mid Y) P(Y)$

Example

Task: Predict whether or not a picnic spot is enjoyable

Training Data: $X = (A_1 \quad A_2 \quad A_3 \quad \dots \quad \dots \quad A_d)$

n rows



Sky	Temp	Humid	Wind	Water	Forecst	EnjoySpt
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

Let's learn $P(Y|A)$ – how many parameters?

Prior: $P(Y = y)$ for all y **$K-1$ if K labels**

Likelihood: $P(A=a|Y = y)$ for all a, y **$(2^d - 1)K$ if d binary features**

Too many parameters to estimate for large d !

Conditional Independence

- Predict Lightning
- From two **conditionally Independent** features
 - Thunder
 - Rain
- parameters needed given L
 - $P(T,R|L)$ **$(2^2-1)2 = 6$**
- With conditional independence assumption
 - $P(T,R|L) = P(T|L) P(R|L)$ **$(2-1)2 + (2-1)2 = 4$**

Naïve Bayes Assumption

- Naïve Bayes assumption:
 - Features are independent given class:

$$P(X_1, X_2|Y) = P(X_1|Y)P(X_2|Y)$$

- More generally:

$$P(X_1 \dots X_d|Y) = \prod_{i=1}^d P(X_i|Y)$$

- How many parameters now? **dK vs. (2^d-1)K**
 - Suppose **X** is composed of *d* binary features

Naïve Bayes Classifier

- Given:
 - Class Prior $P(Y)$
 - d conditionally independent features $\mathbf{X}$ given the class Y
 - For each X_i , we have likelihood $P(X_i|Y)$

- Decision rule:

$$\begin{aligned} f_{NB}(\mathbf{x}) &= \arg \max_y P(x_1, \dots, x_d | y) P(y) \\ &= \arg \max_y \prod_{i=1}^d P(x_i | y) P(y) \end{aligned}$$

- If conditional independence assumption holds, NB is optimal classifier! But worse otherwise.

Naïve Bayes – Discrete features

- Training Data $\{(X^{(j)}, Y^{(j)})\}_{j=1}^n$ $X^{(j)} = (X_1^{(j)}, \dots, X_d^{(j)})$
- NB prediction for the test data: $x = (x_1, \dots, x_d)$
- Maximum Likelihood Estimates

- For Class Prior

$$\hat{P}(y) = \frac{\{\#j : Y^{(j)} = y\}}{n}$$

- For Likelihood

$$\hat{P}(x_i|y) = \frac{\{\#j : X_i^{(j)} = x_i, Y^{(j)} = y\}}{\{\#j : Y^{(j)} = y\}}$$

- NB Prediction for test data

$$Y = \operatorname{argmax}_y \hat{P}(y) \prod_{i=1}^d \hat{P}(x_i|y)$$

Example

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- Class: $P(Y) = N_c/N$

– e.g., $P(\text{No}) = 7/10$,
 $P(\text{Yes}) = 3/10$

- For discrete attributes:

$$P(A_i | Y_k) = |A_{ik}| / N_{kc}$$

- where $|A_{ik}|$ is number of instances having attribute A_i and belongs to class Y_k
- Examples:

$$P(\text{Status}=\text{Married}|\text{No}) = 4/7$$

$$P(\text{Refund}=\text{Yes}|\text{Yes})=0$$

Continuous Attributes?

- For continuous attributes:
 - **Discretize** the range into bins
 - one ordinal attribute per bin
 - violates independence assumption
 - **Two-way split:** $(A < v)$ or $(A > v)$
 - choose only one of the two splits as new attribute
 - **Probability density estimation:**
 - Assume attribute follows a normal distribution
 - Use data to estimate parameters of distribution (e.g., mean and standard deviation)
 - Once probability distribution is known, can use it to estimate the conditional probability $P(A_i|Y)$

How to Estimate Probabilities from Data?

<i>Tid</i>	Refund	Marital Status	Taxable Income	Evade
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- Normal distribution:

$$P(A_i | Y_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} e^{-\frac{(A_i - \mu_{ij})^2}{2\sigma_{ij}^2}}$$

– One for each (A_i, c_i) pair

- For (Income, Class=No):
 - If Class=No
 - sample mean = 110
 - sample variance = 2975

$$P(\text{Income} = 120 | \text{No}) = \frac{1}{\sqrt{2\pi(54.54)}} e^{-\frac{(120-110)^2}{2(2975)}} = 0.0072$$

Example of Naïve Bayes Classifier

$X = (\text{Refund} = \text{No}, \text{Married}, \text{Income} = 120\text{K})$

naive Bayes Classifier:

$$P(\text{Refund}=\text{Yes}|\text{No}) = 3/7$$

$$P(\text{Refund}=\text{No}|\text{No}) = 4/7$$

$$P(\text{Refund}=\text{Yes}|\text{Yes}) = 0$$

$$P(\text{Refund}=\text{No}|\text{Yes}) = 1$$

$$P(\text{Marital Status}=\text{Single}|\text{No}) = 2/7$$

$$P(\text{Marital Status}=\text{Divorced}|\text{No}) = 1/7$$

$$P(\text{Marital Status}=\text{Married}|\text{No}) = 4/7$$

$$P(\text{Marital Status}=\text{Single}|\text{Yes}) = 2/7$$

$$P(\text{Marital Status}=\text{Divorced}|\text{Yes}) = 1/7$$

$$P(\text{Marital Status}=\text{Married}|\text{Yes}) = 0$$

For taxable income:

If class=No: sample mean=110
 sample variance=2975

If class=Yes: sample mean=90
 sample variance=25

$$\begin{aligned} P(X|\text{Class}=\text{No}) &= P(\text{Refund}=\text{No}|\text{Class}=\text{No}) \\ &\quad \times P(\text{Married}|\text{Class}=\text{No}) \\ &\quad \times P(\text{Income}=120\text{K}|\text{Class}=\text{No}) \\ &= 4/7 \times 4/7 \times 0.0072 = 0.0024 \end{aligned}$$

$$\begin{aligned} P(X|\text{Class}=\text{Yes}) &= P(\text{Refund}=\text{No}|\text{Class}=\text{Yes}) \\ &\quad \times P(\text{Married}|\text{Class}=\text{Yes}) \\ &\quad \times P(\text{Income}=120\text{K}|\text{Class}=\text{Yes}) \\ &= 1 \times 0 \times 1.2 \times 10^{-9} = 0 \end{aligned}$$

Since $P(X|\text{No})P(\text{No}) > P(X|\text{Yes})P(\text{Yes})$

Therefore $P(\text{No}|X) > P(\text{Yes}|X)$
 $\Rightarrow \text{Class} = \text{No}$

Naïve Bayes Classifier

- If one of the conditional probability is zero, then the entire expression becomes zero
- Probability estimation:

$$\text{Original : } P(A_i | C) = \frac{N_{ic}}{N_c}$$

c: number of classes

$$\text{Laplace : } P(A_i | C) = \frac{N_{ic} + 1}{N_c + c}$$

p: prior probability

m: parameter

$$\text{m - estimate : } P(A_i | C) = \frac{N_{ic} + mp}{N_c + m}$$

Example of Naïve Bayes Classifier

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

A: attributes

M: mammals

N: non-mammals

$$P(A | M) = \frac{6}{7} \times \frac{6}{7} \times \frac{2}{7} \times \frac{2}{7} = 0.06$$

$$P(A | N) = \frac{1}{13} \times \frac{10}{13} \times \frac{3}{13} \times \frac{4}{13} = 0.0042$$

$$P(A | M)P(M) = 0.06 \times \frac{7}{20} = 0.021$$

$$P(A | N)P(N) = 0.004 \times \frac{13}{20} = 0.0027$$

$$P(A|M)P(M) > P(A|N)P(N)$$

=> Mammals



Naïve Bayes (Summary)

- Robust to isolated noise points
- Handle missing values by ignoring the instance during probability estimate calculations
- Robust to irrelevant attributes
- Independence assumption may not hold for some attributes
 - Use other techniques such as Bayesian Belief Networks (BBN)



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Logistic Regression

Assumes the following functional form for $P(Y|X)$:

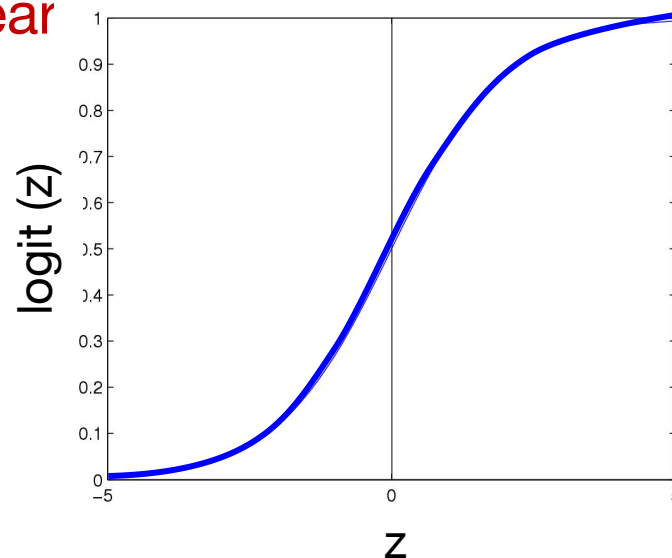
$$P(Y = 1|X) = \frac{\exp(w_0 + \sum_i w_i X_i)}{1 + \exp(w_0 + \sum_i w_i X_i)}$$

Logistic function applied to a linear function of the data

**Logistic function
(or Sigmoid):**

$$\sigma(z) = \frac{\exp(z)}{1 + \exp(z)}$$

$$z = w_0 + \sum_i w_i X_i$$



Features can be discrete or continuous!

A Linear Classifier

Assumes the following functional form for $P(Y|X)$:

$$P(Y = 1|X) = \frac{\exp(w_0 + \sum_i w_i X_i)}{1 + \exp(w_0 + \sum_i w_i X_i)}$$

$$\Rightarrow P(Y = 0|X) = \frac{1}{1 + \exp(w_0 + \sum_i w_i X_i)}$$

$$\Rightarrow \frac{P(Y = 1|X)}{P(Y = 0|X)} = \exp(w_0 + \sum_i w_i X_i) \stackrel{1}{\geq} 0$$

$$\Rightarrow w_0 + \sum_i w_i X_i \stackrel{1}{\geq} 0$$

A Linear Classifier

Assumes the following functional form for $P(Y|X)$:

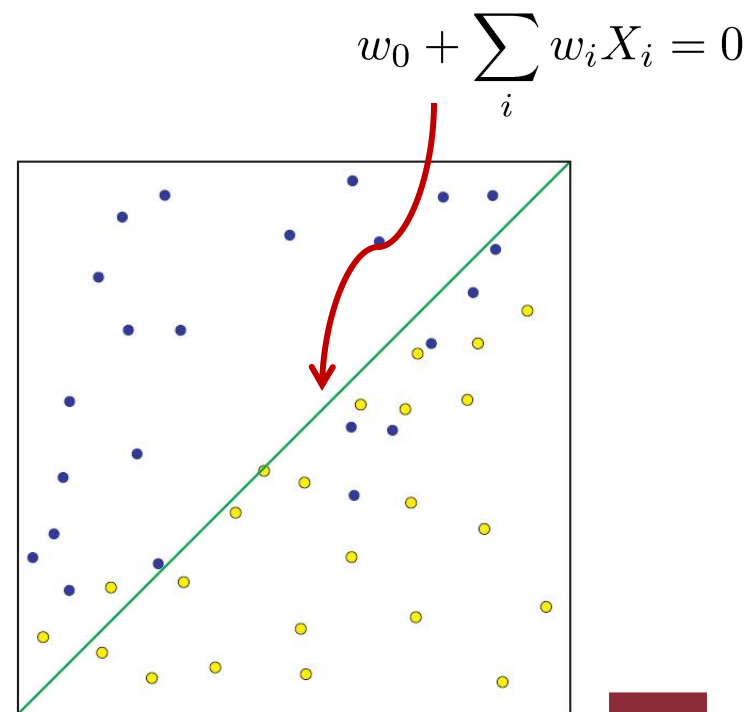
$$P(Y = 1|X) = \frac{\exp(w_0 + \sum_i w_i X_i)}{1 + \exp(w_0 + \sum_i w_i X_i)}$$

Decision boundary:

$$P(Y = 1|X) \underset{0}{\overset{1}{\geq}} P(Y = 0|X)$$

$$w_0 + \sum_i w_i X_i \underset{0}{\overset{1}{\geq}} 0$$

(Linear Decision Boundary)



More than 2 classes?

- Logistic regression in more general case, where

$$Y \in \{y_1, \dots, y_K\}$$

for $k = 1, \dots, K-1$

$$P(Y = y_k|X) = \frac{\exp(w_{k0} + \sum_{i=1}^d w_{ki}X_i)}{1 + \sum_{j=1}^{K-1} \exp(w_{j0} + \sum_{i=1}^d w_{ji}X_i)}$$

for $k = K$ (normalization. so no weights for this class)

$$P(Y = y_K|X) = \frac{1}{1 + \sum_{j=1}^{K-1} \exp(w_{j0} + \sum_{i=1}^d w_{ji}X_i)}$$

Training Logistic Regression

How to learn the parameters $w_0, w_1, \dots, w_d$?

Training Data $\{(X^{(j)}, Y^{(j)})\}_{j=1}^n$ $X^{(j)} = (X_1^{(j)}, \dots, X_d^{(j)})$

Maximum (Conditional) Likelihood Estimates

$$\hat{\mathbf{w}}_{MCLE} = \arg \max_{\mathbf{w}} \prod_{j=1}^n P(Y^{(j)} | X^{(j)}, \mathbf{w})$$

Discriminative philosophy – Don't waste effort learning $P(X)$, focus on $P(Y|X)$ – that's all that matters for classification!

Conditional log Likelihood

$$\left. \begin{aligned} P(Y = 1|\mathbf{X}, \mathbf{w}) &= \frac{\exp(w_0 + \sum_i w_i X_i)}{1 + \exp(w_0 + \sum_i w_i X_i)} \\ P(Y = 0|\mathbf{X}, \mathbf{w}) &= \frac{1}{1 + \exp(w_0 + \sum_i w_i X_i)} \end{aligned} \right\} P(Y = y|\mathbf{X}, \mathbf{w}) = \frac{\exp(y(w_0 + \sum_i w_i X_i))}{1 + \exp(w_0 + \sum_i w_i X_i)}$$

$$\begin{aligned} l(\mathbf{w}) &\equiv \ln \prod_j P(y^{(j)}|\mathbf{x}^{(j)}, \mathbf{w}) \\ &= \sum_j \left[y^{(j)} \left(w_0 + \sum_i^d w_i x_i^{(j)} \right) - \ln \left(1 + \exp \left(w_0 + \sum_i^d w_i x_i^{(j)} \right) \right) \right] \end{aligned}$$

$$\max l(\mathbf{w}) \rightarrow \min -l(\mathbf{w})$$

Maximizing Conditional log Likelihood

$$\begin{aligned}\min_{\mathbf{w}} -l(\mathbf{w}) &\equiv -\ln \prod_j P(y^{(j)} | \mathbf{x}^{(j)}, \mathbf{w}) \\ &= -\sum_j \left[y^{(j)} \left(w_0 + \sum_i^d w_i x_i^{(j)} \right) - \ln \left(1 + \exp \left(w_0 + \sum_i^d w_i x_i^{(j)} \right) \right) \right]\end{aligned}$$

Bad news: no closed-form solution to minimize $-l(\mathbf{w})$

Good news: $-l(\mathbf{w})$ is convex function of $\mathbf{w}$

Optimizing convex function

- Minus Conditional likelihood for Logistic Regression is convex

Gradient Descent (convex)

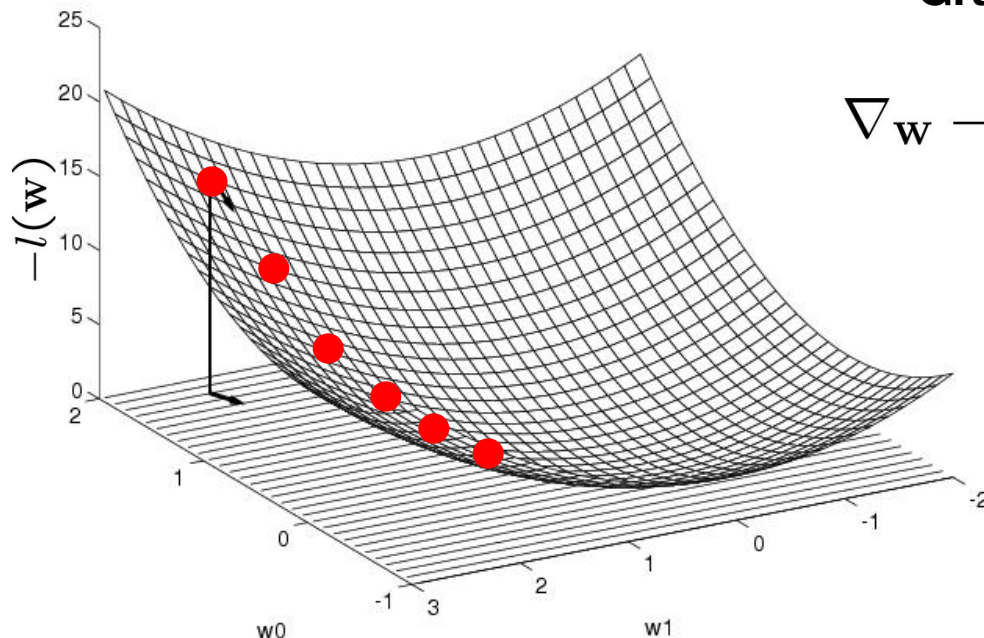
Gradient:

$$\nabla_{\mathbf{w}} -l(\mathbf{w}) = \left[\frac{-\partial l(\mathbf{w})}{\partial w_0}, \dots, \frac{-\partial l(\mathbf{w})}{\partial w_d} \right]$$

Update rule:

Learning rate, $\alpha_t > 0$

$$w_i^{(t+1)} \leftarrow w_i^{(t)} - \alpha_t \frac{-\partial l(\mathbf{w})}{\partial w_i} \Big|_{\mathbf{w}^t}$$



How about MAP?

$$p(\mathbf{w} \mid Y, \mathbf{X}) \propto P(Y \mid \mathbf{X}, \mathbf{w})p(\mathbf{w})$$

- One common approach is to define priors on $\mathbf{w}$
 - Normal distribution, zero mean, identity covariance
- Corresponds to **Regularization**
 - Helps avoid very large weights and overfitting
- M(C)AP estimate

$$\mathbf{w}^* = \arg \max_{\mathbf{w}} \ln \left[p(\mathbf{w}) \prod_{j=1}^n P(y^{(j)} \mid \mathbf{x}^{(j)}, \mathbf{w}) \right]$$

- Regularization

$$\mathbf{w} \sim \mathcal{N}(0, \tau^2 \mathbf{I})$$

$$\mathbf{w}^* = \arg \max_{\mathbf{w}} \ln \left[p(\mathbf{w}) \prod_{j=1}^n P(y^{(j)} | \mathbf{x}^{(j)}, \mathbf{w}) \right] \quad \text{Zero-mean Gaussian prior}$$

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} - \sum_{j=1}^n \ln P(y^{(j)} | \mathbf{x}^{(j)}, \mathbf{w}) + \frac{1}{2\tau^2} \|\mathbf{w}\|^2$$

Penalizes large weights



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Case Study: Spam Email Detection

1. **spam**: Indicator for whether the email was spam.
2. **to_multiple**: Indicator for whether the email was addressed to more than one recipient.
3. **from**: Whether the message was listed as from anyone (this is usually set by default for regular outgoing email).
4. **cc**: Indicator for whether anyone was CCed.
5. **sent_email**: Indicator for whether the sender had been sent an email in the last 30 days.
6. **image**: Indicates whether any images were attached.
7. **attach**: Indicates whether any files were attached.
8. **dollar**: Indicates whether a dollar sign or the word 'dollar' appeared in the email.
9. **winner**: Indicates whether "winner" appeared in the email.
10. **inherit**: Indicates whether "inherit" (or an extension, such as inheritance) appeared in the email.
11. **password**: Indicates whether "password" appeared in the email.
12. **num_char**: The number of characters in the email, in thousands.
13. **line_breaks**: The number of line breaks in the email (does not count text wrapping).
14. **format**: Indicates whether the email was written using HTML (e.g. may have included bolding or active links) or plaintext.
15. **re_subj**: Indicates whether the subject started with "Re:", "RE:", "re:", or "rE":
16. **exclaim_subj**: Indicates whether there was an exclamation point in the subject.
17. **urgent_subj**: Indicates whether the word "urgent" was in the email subject.
18. **exclaim_mess**: The number of exclamation points in the email message.
19. **number**: Factor variable saying whether there was no number, a small number (under 1 million), or a big number.

The End

■ THANK YOU

Any questions?

